KOUSIK KUMAR DUTTA

Ph.D. in Computer Science & Engineering
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About

Ph.D. researcher in Computer Science and Engineering at the Indian Institute of Technology Ropar with research interests in spatial data management, route planning, and scalable graph algorithms. My work focuses on developing efficient algorithms and systems for constrained optimization in large time-dependent networks and intelligent transportation systems. I aim to build scalable and intelligent graph-processing solutions by integrating algorithmic techniques, data systems, and high-performance computing approaches.

Research Interests

- **Spatial Data Management:** Efficient storage, querying, and processing of large-scale spatio-temporal and road network data.
- **Route Planning and Network Optimization:** Preference-aware and constrained optimization methods for path computation in dynamic transportation networks.
- **Graph Algorithms for Large-Scale Networks:** Designing scalable algorithms for shortest-path and graph-search problems in large graphs.
- **Intelligent Transportation Systems:** Data-driven approaches for mobility analytics, navigation systems, and transportation optimization.
- **Scalable Graph Processing and Analytics:** GPU-accelerated, secondary-memory, and learning-guided techniques for large graph computation.

Education

Ph.D. in Computer Science and Engineering *2021–2026 Indian Institute of Technology Ropar, Punjab, India*

- **Dissertation:** "Constrained Maximization of Preferences on Time-Dependent Road Networks"
- **Advisor:** Dr. Venkata. M. V. Gunturi & Venkata Kalyan Tavva
- **CGPA:** 7.92/10.00

M.Tech in Computer Science and Engineering *2019–2021 Indian Institute of Technology Ropar, Punjab, India*

- **Thesis:** "A Multi-Threading Algorithm for Constrained Path Optimization on Road Networks"
- **CGPA:** 8.39/10.00

B.Tech in Computer Science and Engineering *2014–2018 Kalyani Government Engineering College, West Bengal, India*

- **CGPA:** 7.45/10.00

Publications

Journal Publications

[J1] **Kousik Kumar Dutta**, Venkata M. V. Gunturi. A Parallelizable Algorithm for Constrained Maximization of Preferences in Large Time-Dependent Graphs. *Geoinformatica*, vol. 30, article 19, 2026. DOI:

https://doi.org/10.1007/s10707-026-00577-z

Refereed Conference Publications

[C1] **Kousik Kumar Dutta**, Venkata M. V. Gunturi. A User-Configurable Navigation System for Wideness and Turn-Aware Routing. *International Conference on Mobile Data Management (MDM)*, 2026. Accepted.

[C2] **Kousik Kumar Dutta**, Venkata M. V. Gunturi. Interval-Based Constrained Path Optimization in Time-Dependent Road Networks. *26th International Conference on Web Information Systems Engineering (WISE)*, 2025. DOI: https://doi.org/10.1007/978-981-95-7251-9_12

[C3] **Kousik Kumar Dutta**, Anirban Dewan, Venkata M. V. Gunturi. A Multi-Threading Algorithm for Constrained Path Optimization on Road Networks. *23rd International Conference on Web Information Systems Engineering (WISE)*, 2022. DOI: https://doi.org/10.1007/978-3-031-20891-1_9

[C4] **Kousik Kumar Dutta**, Prathamesh N. Tanksale, Shirshendu Das. A Fairness Conscious Cache Replacement Policy for Last Level Cache. *Design, Automation and Test in Europe (DATE)*, 2021. DOI: https://doi.org/10.23919/DATE51398.2021.9474096

[C5] Ankita Dewan, Venkata M. V. Gunturi, Vinayak Naik, **Kousik Kumar Dutta**. NEAT Activity Detection Using Smartwatch at Low Sampling Frequency. *IEEE SmartWorld/SCALCOM/UIC/ATC/IOP/SCI*, 2021. DOI: https://doi.org/10.1109/SWC50871.2021.00014

Manuscripts Under Review

[U1] **Kousik Kumar Dutta**, Venkata M. V. Gunturi. A Bi-Directional Search for Interval Constrained Path Optimization in Large Time-Dependent Road Networks. Submitted to *Knowledge and Information Systems*, 2026. (Under review)

[U2] **Kousik Kumar Dutta**, Venkata M. V. Gunturi. Streamlining Vehicle Logistics: Maximizing Pickup–Delivery under Non-Additive Loading–Unloading Costs. Submitted to *Journal of Scheduling*, 2026. (Under review)

Workshop Publications

[W1] **Kousik Kumar Dutta**, Venkata M. V. Gunturi. Constrained Path Optimization on Time-Dependent Road Networks. *WISE PhD Symposium, Demos and Workshops*, 2024. DOI: https://doi.org/10.1007/978-981-96-1483-7

[W2] Prathamesh N. Tanksale, **Kousik Kumar Dutta**, Shirshendu Das, T. V. Kalyan. CAMOUFLAGE: An Efficient Mechanism to Hide Congestion in NVM LLC. Extended Abstract, *IEEE International Conference on High Performance Computing (HiPC)*, 2023.

Current Research Projects

Learning-Guided Search Optimization for Large Road Networks

- Investigating reinforcement learning-based approaches for reducing the search space in constrained route optimization problems on large road networks.
- Exploring adaptive search policies that dynamically prioritize candidate paths based on graph topology, temporal characteristics, and user preferences.
- Studying the integration of learning-guided strategies with conventional graph search techniques to improve scalability while maintaining solution quality.

GPU-Accelerated Graph Algorithms for Large Road Networks

- Investigating scalable graph processing frameworks for large road networks using GPU architectures.
- Exploring parallel graph traversal and shortest-path computation techniques while addressing memory hierarchy constraints and irregular graph structures.
- Studying optimization strategies involving memory coalescing, workload balancing, and minimizing warp divergence for graph workloads.

Database and Secondary-Memory Optimization for Graph Processing

- Investigating external-memory and SQL-based approaches for scalable processing of large graph datasets.

- Exploring index structures and storage-aware algorithms for graph query processing under memory constraints.
- Studying efficient graph representations and query execution strategies for large-scale spatio-temporal networks.

Academic Services

Teaching Assistantships — Indian Institute of Technology Ropar

- **Database Management Systems (B.Tech. Core Course)**
  - Conducted and evaluated laboratory assignments involving SQL, relational database design, and query processing.
  - Mentored student teams in designing and implementing course projects including Academic Database Management Systems and Railway Reservation Systems.
- **Software Engineering (B.Tech. Core Course)**
  - Evaluated laboratory assignments and guided students in software development practices.
  - Assisted student teams in developing software systems from requirement analysis to implementation, including Software Requirement Specification (SRS) design, architectural decisions, design pattern selection, unit testing, and software development workflow.
- **Postgraduate Software Laboratory (M.Tech. Core Course)**
  - Mentored postgraduate students in implementing advanced data structures and algorithmic techniques.
  - Guided projects involving large-scale graph processing and secondary-memory algorithms, including external-memory Dijkstra implementations and Extendable Hashing.

Academic Reviewer

- Reviewer for *Geoinformatica* (2023–Present); completed peer review of three manuscripts in areas including spatial databases, graph processing, and route planning.

Awards, Fellowships, and Grants

- **Institute Fellowship (Ph.D.):** Awarded full institute fellowship for doctoral studies at Indian Institute of Technology Ropar (2021–2026).
- **MHRD GATE Scholarship (M.Tech.):** Awarded Ministry of Human Resource Development (MHRD) scholarship during M.Tech studies at Indian Institute of Technology Ropar (2019–2021).
- **GATE 2019:** Secured All India Rank (AIR) 644 in Computer Science and Engineering.

Technical Skills

- **Programming Languages:** C, C++, Java, Python, SQL, CUDA
- **Database Systems:** PostgreSQL, MySQL, Db2
- **Computing Platforms and Libraries:** CUDA Toolkit, OpenMP
- **Development Tools:** Git, LaTeX, VS Code, Eclipse, Microsoft Office Suite

Invited Talks and Presentations

- Presenter at 2026 International Conference on Mobile Data Management (MDM) on User-Configurable Navigation Systems
- Presenter at 26th International Conference on Web Information Systems Engineering (WISE 2025) on Interval-Based Constrained Path Optimization
- Presented PhD research at WISE 2024 PhD Symposium and Demos/Workshops

References

1. **Dr. Venkata M. V. Gunturi** Email: V.Gunturi@lboro.ac.uk
   Lecturer (Assistant Professor), Department of Computer Science
   Loughborough University, United Kingdom
   Relationship: Ph.D. Advisor
   Web: https://www.lboro.ac.uk/departments/compsci/staff/venkata-maruti-viswanath-gunturi/

2. **Dr. T. V. Kalyan** Email: kalyantv@iitrpr.ac.in
   Associate Professor, Department of Computer Science and Engineering
   Indian Institute of Technology Ropar, India
   Relationship: Research Collaborator / M.Tech Mentor
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3. **Dr. Shirshendu Das** Email: shirshendu@cse.iith.ac.in
   Associate Professor, Department of Computer Science and Engineering
   Indian Institute of Technology Ropar, India
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